

Questions on two Sample t-test

June 2023 (done in class)

1. Find the value of t-test for following set of values (4 marks)
~~7, 2, 9, 8~~ 7, 2, 9, 8 and 1, 2, 3, 4

Soln: To perform a t-test, we need to decide what kind of t-test to use. Since you have provided two sets of values we can perform independent sample t-test

Let's go step-by-step to find the t-test

Data

Group 1: 7 2 9 8

Group 2: 1 2 3 4

Step 1: State the null hypothesis

- NULL hypothesis (H_0): There is no significant difference between the means of the two groups i.e. $\mu_1 = \mu_2$
- Alternate hypothesis (H_1): There is a significant difference between the means of two groups i.e. $\mu_1 \neq \mu_2$

Step 2: Calculate the mean of each group

Group 1: mean of Group 1

$$\bar{x}_1 = \frac{7+2+9+8}{4} = \frac{26}{4} = 6.5$$

Group 2: mean of Group 2

$$\bar{x}_2 = \frac{1+2+3+4}{4} = \frac{10}{4} = 2.5$$

p 3: Calculate the variance of each group

Variance formula

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n-1}$$

Group 1 Variance

$$= \frac{(7-6.5)^2 + (2-6.5)^2 + (9-6.5)^2 + (8-6.5)^2}{4-1}$$

$$= \frac{(0.5)^2 + (-4.5)^2 + (2.5)^2 + (1.5)^2}{3}$$

$$= \frac{0.25 + 20.25 + 6.25 + 2.25}{3}$$

$$= \frac{29}{3}$$

$$= 9.67$$

Group 2 Variance

$$= \frac{(1-2.5)^2 + (2-2.5)^2 + (3-2.5)^2 + (4-2.5)^2}{4-1}$$

$$= \frac{(-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2}{3}$$

$$= \frac{2.25 + 0.25 + 0.25 + 2.25}{3}$$

$$= \frac{5}{3}$$

$$= 1.66$$

Step 4: Calculate the pooled Standard deviation

The pooled standard deviation S_p is used because the sample sizes are equal

$$S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

$$= \sqrt{\frac{(4-1)(9.67) + (4-1)(1.67)}{4+4-2}}$$

$$= \sqrt{\frac{3(9.67) + 3(1.67)}{6}}$$

$$= \sqrt{\frac{29.01 + 5.01}{6}}$$

$$= \sqrt{\frac{34.02}{6}}$$

$$= \sqrt{5.67}$$

$$= 2.38$$

Step 5: Calculate the t-statistics

the formula for t-statistics is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{6.5 - 2.5}{2.38 \sqrt{\frac{1}{4} + \frac{1}{4}}}$$

$$= \frac{4.0}{2.38 \sqrt{0.5}} = \frac{4.0}{2.38 \times 0.707} = \frac{4.0}{1.68} = 2.38$$

Degrees of freedom and critical value

Degrees of freedom (df) for an independent samples t-test is,

$$\begin{aligned}df &= n_1 + n_2 - 2 \\&= 4 + 4 - 2 \\&= 6\end{aligned}$$

Using a t-distribution table for 6 degrees of freedom, significance level $\alpha = 0.05$ for a two-tailed test, the critical value is approximately ± 2.447 .

Decision

$|t| > \text{critical value}$, reject H_0

$t = 2.38$ and critical value is 2.447

$|t| = 2.38$, we fail to reject the null hypothesis

Conclusion

There is not enough evidence to suggest a significant difference between the means of two groups at 0.05 significance level.

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Final test marks of the students are as follows (7 marks)

5, 9, 7, 12, 8.5, 6.5, 8, 11, 9.9

Population mean score is 12 and standard deviation is 1.80.

Is the mean value for student significantly different from

population mean value? Let alpha value is 0.05. So

corresponding t-value is 2.62

Given Data

Final test marks of students: 10.5, 9, 7, 12, 8.5, 6.5, 8, 11, 9.9

Population mean score (μ) = 12

Population standard deviation (σ) = 1.80

Significance level (α) = 0.05

Critical t-value = 2.262

Step 1: State the hypothesis

- Null hypothesis (H_0): The average number of days a person spends on vacation is $\mu = 5$ days
- Alternate hypothesis (H_1): The average number of days a person spends on vacation is greater than 5 days ($\mu > 5$ days)

This is one-tailed right test

Step 2: Calculate test-statistics value (t-value)

The formula for t-statistics is

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

where

- $\bar{x} = 9$ (Sample mean)
- $\mu = 5$ (population mean)
- $s = 3$ (Sample Standard deviation)
- $n = 16$ (Sample size)

$$t = \frac{9 - 5}{\frac{3}{\sqrt{16}}}$$

$$= \frac{4}{\frac{3}{4}} = \frac{4}{0.75} = 5.33$$

$$= \frac{16}{3}$$

$$= 5.33$$

Step 3: Compare the t-value with Critical value

- The Calculated t-value is 5.33
- The Critical t-value is 1.753 for a one-tailed test at a significance level of 0.05 and degrees of freedom ($n-1=15$)

Step 5: Compare t-value with Critical value

The calculated t-value is -4.58 and critical t-value is 2.262 at a significance level of 0.05 for a two-tailed test

Step 6: Decision Rule

If $|t| > 2.262$, reject the null hypothesis

$|t| < 2.262$, fail to reject the null hypothesis

Since $|t| = 4.58$ is greater than 2.262 , we reject null hypothesis

Step 7: Conclusion

There is sufficient evidence to conclude that the mean score of students is significantly different from population mean at 0.05 level of significance level

June 2023

(Cmant)

3) A sample size of 16 people is taken. Suppose a claim is made that the average number of days a person spends a vacation is more than equal to 5 days based on sample of 16 people whose mean come out to be 9 days. Sample standard deviation is found to be 3 days at a level of significance of 0.05 and t-value right-tailed test comes out to be 1.753 . Calculate the value of t and check null hypothesis

Solⁿ:

Given Data

- Sample size $n = 16$
- Sample mean $(\bar{x}) = 9$ days
- Hypothesized mean $(\mu) = 5$ days
- Sample standard deviation $(s) = 3$ days
- Significance level $(\alpha) = 0.05$
- Critical t-value for right-tailed test = 1.753

- $(8 - 9.46)^2 = 2.1316$
- $(11 - 9.46)^2 = 2.3716$
- $(9.9 - 9.46)^2 = 0.1936$

Now sum them up

$$\sum (x_i - \bar{x})^2 = 1.1025 + 0.2116 + 0.6576 + 6.4516 + 0.9216 + 8.7501 + 2.1316 + 2.3716 + 0.1936 +$$

$$= 22.1978$$

Now divide by $n-1$

$$= 9 - 1 = 8$$

$$S^2 = \frac{22.1978}{8}$$

$$S^2 = 2.7747$$

$$S = \sqrt{2.7747}$$

$$S = 1.665$$

Step 4: Calculate the Test statistics (t-value)

The formula for test-statistics is

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}$$

Where

$$\bar{x} = 9.46 \text{ (sample mean)}$$

$$\mu = 12 \text{ (population mean)}$$

$$S = 1.665 \text{ (sample standard deviation)}$$

$$n = 9 \text{ (sample size)}$$

Substitute the values

$$t = \frac{9.46 - 12}{\frac{1.665}{\sqrt{9}}}$$

$$= \frac{-2.54}{\frac{1.665}{3}} \approx -4.58$$

We need to check if the mean value of students significantly differ from ^{mean} population mean-value using a t-test

Step 1: State the hypothesis

- NULL hypothesis (H_0): The mean of the student's score is equal to the population mean, $\mu = 12$
- Alternative hypothesis (H_1): The mean of the student's score is different from population mean, $\mu \neq 12$

Step 2: Calculate the Sample mean

We need to calculate the sample mean $\bar{x}$ of the given student's marks

$$\bar{x} = \frac{\sum x_i}{n} = \frac{10.5 + 9 + 9.7 + 12 + 8.5 + 6.5 + 8 + 11 + 9.9}{9}$$

$$\bar{x} = \frac{85.1}{9}$$

$$\bar{x} = 9.46$$

So, the sample mean is approximately 9.46

Step 3: Calculate Sample Standard deviation

Let's calculate sample standard deviation (s) using the formula

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

We use $n-1$ for sample standard deviation

First, calculate $(x_i - \bar{x})^2$ for each value {when you are working w. pop. standard deviation then use n }

$$\bullet (10.5 - 9.46)^2 = 1.1025$$

$$\bullet (9 - 9.46)^2 = 0.2116$$

$$\bullet (9.7 - 9.46)^2 = 0.0576$$

$$\bullet (12 - 9.46)^2 = 6.4516$$

$$\bullet (8.5 - 9.46)^2 = 0.9216$$

$$\bullet (6.5 - 9.46)^2 = 8.7516$$

Step 7: Conclusion

There is significant difference between the marks of boys and girls at 0.05 significance level.

Chi-Square test

Dec 2023

- 1) Calculate the chi-square value for the following data of incidences of Water borne diseases in three tropical regions (8 marks)

	India	Equador	South America	Total
Typhoid	31	14	45	90
cholera	2	5	53	60
Diarrhoea	53	45	2	100
	86	64	100	250

Soln:

Step 1: State the hypothesis

- NULL hypothesis: There is no significant association between (H₀) the tropical regions and the indices of Water borne diseases.
- Alternate hypothesis: There is a significant association between the tropical regions and the indices of Water borne diseases.

Step 2: Calculate the Expected frequencies

The expected frequency for each cell is calculated using formula

$$E = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

The formula for the t-test is

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

where

$$\bar{X}_1 = 9.78$$

$$\bar{X}_2 = 15.1$$

$$S_1 = 4.12$$

$$S_2 = 4.28$$

$$n_1 = 9 \text{ (number of boys)}$$

$$n_2 = 10 \text{ (number of girls)}$$

$$t = \frac{9.78 - 15.1}{\sqrt{\frac{4.12^2}{9} + \frac{4.28^2}{10}}}$$

$$= \frac{-5.32}{\sqrt{1.88 + 1.83}}$$

$$= \frac{-5.32}{\sqrt{3.71}} = \frac{-5.32}{1.93} \approx -2.76$$

Step 5: Compare the Calculated t-value with the Critical t-value

Degrees of freedom (df): For two sample t-test, we use smaller $n_1 - 1$ and $n_2 - 1$. Thus $df = 8$

Critical t-value (from the question) 2.11 for $\alpha = 0.05$ (two-tail)

Step 6: Decision Rule

If $|t| > 2.11$, reject the null hypothesis

Since the Calculated $t = -2.76$ and $|t| = 2.76 > 2.11$, we reject null hypothesis

Step 2: Calculate the means of both groups

Mean of boys marks ($\bar{x}_1$)

$$\bar{x}_1 = \frac{12+14+10+8+16+5+3+9+11}{9}$$

$$= \frac{88}{9}$$

$$\approx 9.78$$

Mean of girls marks $\bar{x}_2$

$$\bar{x}_2 = \frac{21+18+14+20+11+19+8+12+13+15}{10}$$

$$= \frac{151}{10}$$

$$\approx 15.1$$

Step 3: Calculate standard deviation of both groups

Standard deviation of boy's group

$$S_1 = \sqrt{\frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1}}$$

$$S_1 = \sqrt{\frac{(12-9.78)^2 + (14-9.78)^2 + (10-9.78)^2 + (16-9.78)^2 + (5-9.78)^2 + (3-9.78)^2 + (9-9.78)^2 + (11-9.78)^2}{8}}$$

$$S_1 = 4.12$$

Standard deviation of girls group

$$S_2 = \sqrt{\frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1}}$$

$$S_2 \approx 4.28$$

Step 4: Decision Rule

- If $t > 1.753$, reject the null hypothesis H_0
- If $t \leq 1.753$, fail to reject the null hypothesis H_0

Since $t = 5.33$ is greater than 1.753 , we reject the null hypothesis

Step 5: Conclusion

There is sufficient evidence at the 0.05 significance level to conclude that average number of days a person spends on vacation is significantly greater than 5 days

June 2023

4) The marks ~~for~~ of boys and girls are as follows (6 marks)

Boys: 12, 14, 10, 8, 16, 5, 3, 9, 11

girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 15

Is there any significant difference between the marks of girls and boys? Assume $\alpha = 0.05$ t -value (critical) = 2.11

Soln:

Step 1: State the hypothesis

- Null hypothesis (H_0): There is no significant difference between the marks of boys and girls ($\mu_1 = \mu_2$)
- Alternate hypothesis (H_1): There is significant difference between marks of boys and girls ($\mu_1 \neq \mu_2$)

This is two-tailed test

Numerical on the Concept of ANOVA

1) To assess the significance of possible variation in a certain test between Convent Schools of city, a common test was given to the number of students taken at a random from the fifth class of 3 schools concerned, the results are given below $F_{0.05} = 3.89$

(8 marks)

A	B	C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Make Analysis of Variance for the given data

Step 1: Define the hypothesis

Null hypothesis (H_0): There is no significant difference in means of test score between the three schools

Alternate hypothesis (H_1): At least one school's mean score is significantly different from the others

Step 2: Calculate the mean for each group

• Mean of A ($\bar{X}_1$):

$$\bar{X}_1 = \frac{9+11+13+9+8}{5}$$

$$= \frac{50}{5}$$

$$\bar{X}_1 = 10$$

Step 4: Sum the chi-square values

$$\chi^2 = 0.006052 + 3.54 + 2.26 + 17.10 + 7.02 + 35.94 + 19.06 + 19.82 + 36.11$$
$$= 125.96$$

Step 5: determine the Degrees of freedom

The degrees of freedom (df) are calculated as

$$df = (\text{number of rows} - 1) \times (\text{number of columns} - 1)$$

In this case there are 3 rows and 3 columns so

$$df = (3 - 1) \times (3 - 1)$$
$$= 2 \times 2$$
$$= 4$$

Step 6: Compare the chi-square value with critical value

To determine, if the result is significant, we compare the calculated chi-square value with the critical value from the chi-square distribution table at a significance level of 0.05 for 4 degrees of freedom.

The critical value is approximately 9.488

Since the calculated value i.e. χ^2 value = 125.96 is greater than critical value we reject the null hypothesis.

$\therefore$ there is significant association between the tropical and incidence of water borne diseases.

Step 3: Compare chi-Square Statistics

The chi-square formula is

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

where O is observed value and E is the Expected value

For typhoid

$$\bullet \frac{(31-30.96)^2}{30.96} = \frac{0.0016}{30.96} = 0.000052$$

$$\bullet \frac{(14-23.04)^2}{23.04} = \frac{81.5376}{23.04} = 3.54$$

$$\bullet \frac{(45-36.00)^2}{36.00} = \frac{81.00}{36.00} = 2.25$$

For cholera

$$\bullet \frac{(2-20.64)^2}{20.64} = \frac{352.9216}{20.64} = 17.10$$

$$\bullet \frac{(5-15.36)^2}{15.36} = \frac{107.7696}{15.36} = 7.02$$

$$\bullet \frac{(53-24.00)^2}{24.00} = \frac{841.00}{24.00} = 35.04$$

For diarrhoea

$$\bullet \frac{(53-34.40)^2}{34.40} = \frac{345.96}{34.40} = 10.06$$

$$\bullet \frac{(45-25.60)^2}{25.60} = \frac{380.25}{25.60} = 14.85$$

$$\bullet \frac{(2-40.00)^2}{40.00} = \frac{1444.00}{40.00} = 36.10$$

Let's calculate the expected values for each cell in the table

For typhoid

$$E_{11}(\text{India}) = \frac{90 \times 86}{250} \\ = 30.96$$

$$E_{12}(\text{Ecuador}) = \frac{90 \times 64}{250} \\ = 23.04$$

$$E_{13}(\text{South America}) = \frac{90 \times 100}{250} \\ = 36.00$$

For Cholera

$$E_{21}(\text{India}) = \frac{60 \times 86}{250} \\ = 20.64$$

$$E_{22}(\text{Ecuador}) = \frac{60 \times 64}{250} \\ = 15.36$$

$$E_{23}(\text{South America}) = \frac{60 \times 100}{250} \\ = 24.00$$

For diarrhoea

$$E_{31}(\text{India}) = \frac{100 \times 86}{250} \\ = 34.40$$

$$E_{32}(\text{Ecuador}) = \frac{100 \times 64}{250} \\ = 25.60$$

$$E_{33}(\text{South America}) = \frac{100 \times 100}{250} \\ = 40.40$$

8: Calculate F-statistics

$$F = \frac{MSB}{MSW}$$

$$= \frac{5}{11.5}$$

$$= 0.435$$

9: Compare F-Statistics with critical value

The critical value of $F_{0.05}$ is given as 3.89

Since the Calculated F-value is (0.435) is less than critical value (3.89) we fail to reject null hypothesis

Conclusion

There is no significant difference ^{in the test scores} between the three schools at the 5% significance level

$$\text{Total SSW} = \text{SSW}_A + \text{SSW}_B + \text{SSW}_C$$

$$= 16 + 58 + 64$$

$$= 138$$

Step 6: Calculate the degrees of freedom

- Degrees of freedom between groups (df_1)

$$k - 1 = 3 - 1$$

$$= 2$$

- Degrees of freedom within groups (df_2)

$$N - k = 15 - 3$$

$$= 12$$

where $k = 3$ (Number of groups)

$N = 15$ (Total number of observations)

Step 7: Calculate mean square between groups (MSB) and mean square within groups (MSW)

- mean square between groups

$$MSB = \frac{SSB}{df_1}$$

$$= \frac{10}{2}$$

$$= 5$$

- Mean square within groups

$$MSW = \frac{SSW}{df_2}$$

$$= \frac{138}{12} = 11.5$$

$$SSB = 5(10-11)^2 + 5(11-11)^2 + 5(12-11)^2$$

$$SSB = 5(1)^2 + 5(0) + 5(1)$$

$$SSB = 5 + 5$$

$$SSB = 10$$

Step 5: Calculate the sum of squares within groups (SSW)

The formula for SSW is

$$SSW = \sum (x_{ij} - \bar{x}_i)^2$$

where x_{ij} are individual observations, and $\bar{x}_i$ is the mean of group

For group A

$$SSW_A = (9-10)^2 + (11-10)^2 + (13-10)^2 + (9-10)^2 + (8-10)^2$$

$$SSW_A = 1 + 1 + 9 + 1 + 4$$

$$SSW_A = 16$$

For group B

$$SSW_B = (13-11)^2 + (12-11)^2 + (10-11)^2 + (15-11)^2 + (5-11)^2$$

$$SSW_B = 4 + 1 + 1 + 16 + 36$$

$$= 58$$

For group C

$$SSW_C = (14-12)^2 + (13-12)^2 + (17-12)^2 + (7-12)^2 + (9-12)^2$$

$$SSW_C = 4 + 1 + 25 + 25 + 9$$

$$= 64$$

• Mean of B ($\bar{X}_2$):

$$\bar{X}_2 = \frac{13+12+10+15+5}{5}$$

$$= \frac{55}{5}$$

$$= 11$$

• Mean of C ($\bar{X}_3$):

$$\bar{X}_3 = \frac{14+13+17+7+9}{5}$$

$$= \frac{60}{5}$$

$$= 12$$

Step 3: Calculate overall Mean (Grand mean)

$$\bar{X} = \frac{(9+11+13+9+8+13+12+10+15+5+14+13+17+7+9)}{15}$$

$$= \frac{165}{15}$$

$$= 11$$

Step 4: Calculate Sum of Squares between groups (SSB)

$$SSB = n_1(\bar{X}_1 - \bar{X})^2 + n_2(\bar{X}_2 - \bar{X})^2 + n_3(\bar{X}_3 - \bar{X})^2$$

$$n_1 = n_2 = n_3 = 5 \text{ (number of observations in each group)}$$

$$\bar{X}_1 = 10, \bar{X}_2 = 11, \bar{X}_3 = 12 \text{ and } \bar{X} = 11$$